# Ejercicio 4_5

CKs
APs
PK

3FN

TAREA 7

**Ordenes**

| A | B | C | D | E | F | G | H | I |
|---|---|---|---|---|---|---|---|---|
| Id_orden | Fecha_or | Id_cliente | Nom_cliente | Estado | Num_art | nom_art | cant_art | Precio_art |
| 2301 | 23/02/11 | 101 | Martin | Caracas | 3786 | Red | 3 | 35,00 |
| 2301 | 23/02/11 | 101 | Martin | Caracas | 4011 | Raqueta | 6 | 65,00 |
| 2301 | 23/02/11 | 101 | Martin | Caracas | 9132 | Paq-3 | 8 | 4,75 |
| 2302 | 25/02/11 | 107 | Herman | Coro | 5794 | Paq-6 | 4 | 5,00 |
| 2303 | 27/02/11 | 110 | Pedro | Maracay | 4011 | Raqueta | 2 | 65,00 |
| 2303 | 27/02/11 | 110 | Pedro | Maracay | 3141 | Funda | 2 | 10,00 |

orden — cliente — articulo

$A \rightarrow B, C$

$C \rightarrow D, E$

$F \rightarrow G, I$

$\{A, F\} \rightarrow H$

atributos dependientes: $\{B, C, D, E, G, I, H\}$

Atributos Independientes: $\{A, F\}$

como A y F no puede ser determinado por nadiemás:

$AF^+ = \{A, F, B, C, G, I, D, E, H\}$ (CK) (PK)

$APs = [A, F]$

$AnPs = [B, C, D, E, G, I, H]$

2NF pide que no haya atributos que dependan de solo una parte de PK

A → B, C, D, E ✗

F → G, I ✗

A, F → H ✓

| Ordenes_info | | | | |
|---|---|---|---|---|
| A | B | C | D | E |

| Articulos | | |
|---|---|---|
| F | G | I |

| Detalle_orden | | |
|---|---|---|
| A | F | H |

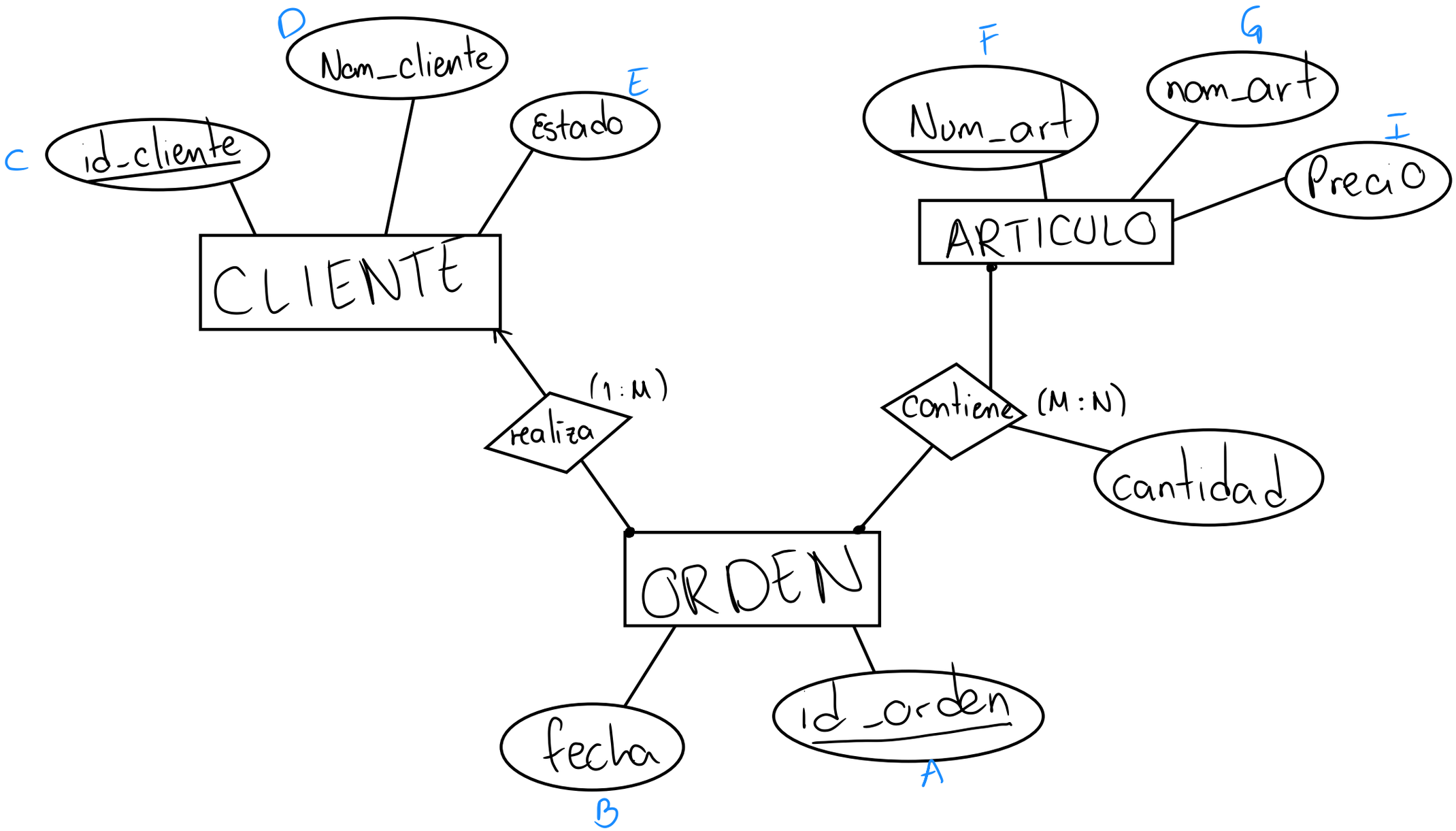
3NF pide que los atributos que no son llave, dependan solo de la PK

en Ordenes_info → A → C → D, E transitivo

| Orden | | |
|---|---|---|
| A | B | C |

| Cliente | | |
|---|---|---|
| C | D | E |

| Articulos | | |
|---|---|---|
| F | G | I |

| Detalle_orden | | |
|---|---|---|
| A | F | H |

# Modelo Entidad Relación